

Reference excerpt 01-004

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License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The note contains an original reasoning of mine and the goal to share thoughts and methodologies, not results. Therefore before using the contents of these notes, everyone is invited to verify the accuracy of the assumptions and conclusions. BACKGROUND The brain drainage is due to the venous blood flow directed from the brain to the heart through the internal jugular veins (IJVs), epidural veins and vertebral veins (see Fig. 1). The brain-heart direction indicates that there must be a negative pressure gradient driving the blood flow (i.e. the pressure in the brain is higher than the pressure in the right atrium (RA)). The pressure, where the IJVs begin, is the residual arterial pressure and, despite its pulsatile nature, in this study it is considered to be constant. The pressure in the RA varies according to the cardiac cycle. Its trace presents two wave peaks called a and v and two waves minima called x and y. Such waves have a precise phase relationship with the ECG waves PQRS [Applefeld(1990)], see Fig. 2. The waves a, x, v and y are also detectable at level of the neck due to the internal jugular vein (IJV) pulsation [Mackenzie(1902)]. The pressure changes generated in the RA are indeed transmitted to the IJV and affect the velocity of the blood in two ways i) modifying the pressure gradient so that it is no longer constant over time and